Food Policy 84 (2019) 111–120

Contents lists available at ScienceDirect

Food Policy

journal homepage: www.elsevier.com/locate/foodpol

The implications of heterogeneous habit in consumer beverage purchases on soda and sin taxes☆

Wenying Li, Jeffrey H. Dorfman*

Department of Agricultural and Applied Economics at The University of Georgia, United States

ARTICLE INFO

Keywords:
Heterogeneity
Habit strength
Sin tax
Soda tax
Smooth transition

ABSTRACT

In this article, we study the heterogeneity of habit strength in households' demand for regular carbonated sweetened beverages (CSBs) and beer in the United States. A demand model that nests a smooth transition function is used to describe habit-based consumption patterns, revealing heterogeneous strengths of habits among households. We find that more habitual consumers, those with a strong preference for a particular product, are not as sensitive to price or expenditure as the aggregate population. This finding is further supported by simulations of the potential effects of soda and beer taxes. We find the aggregate response to soda and beer taxes is smaller than when the influence of habit is assumed to be homogeneous.

1. Introduction

There are numerous studies that analyze consumer responses and welfare effects due to government policies related to products with habit effects. This type of research is specifically targeted at unhealthy and habit-forming products, such as sugary beverages, junk food, alcohol, and tobacco, etc. (Andreyeva et al., 2010; Goryakin et al., 2017; Härkänen et al., 2014; Lin et al., 2017; Pacula, 1998). The models employed to study such policies consider price and income as the main determinants of consumption. A general question is whether increasing the cost would, in fact, decrease the consumption of unhealthy products and break consumers' original habits, thus improving health outcomes. To answer such questions, a common practice is to estimate an econometric model for consumer demand under habit formation. Then, the associated economic impact under different tax or subsidy policies is simulated with the estimated model to assess such policies (Duffy, 2003; Haden, 1990; Herzfeld et al., 2014; Zhen et al., 2014, 2011; Zheng et al., 2017).

Over decades, demand models have evolved to better describe people's behavior as well as to be consistent with economic theory. Models have been greatly improved by considering nonlinear income effects, endogeneity, dynamics, and censoring, etc. But one strict econometric assumption, homogeneity, has rarely been relaxed. Andreyeva et al. (2010) reviewed all US-based studies on the price elasticities of demand for major food categories. Most of these studies obtain price elasticity estimates from traditional homogeneous models. By assuming all consumers react equally to changes, such models might yield biased estimates of aggregate responses if heterogeneous responses are not symmetrically distributed. Moreover, intuition tells us there exists extreme variation in preference over specific goods, such as tobacco, alcohol, and sweetened beverages, which severely undermines the assumption of homogeneous responses. In the last decade, incorporating consumer heterogeneity in demand models has received increasing attention. The most prominent examples are Bertail and Caillavet (2008) and Wang (2015). Bertail and Caillavet (2008) studied the heterogeneity of fruit and vegetable consumption patterns in France. A finite mixture of AIDS models is used to describe food demand patterns revealing different preferences. Six different clusters which reflect specific socio-demographic characteristics and different income and price elasticities are obtained. Wang (2015) provides estimates of the relevant price elasticities based on a random coefficient dynamic demand model that addresses unobservable persistent heterogeneous tastes. It is found that traditional static analyses tend to overestimate the long-run own-price elasticity of regular soda by 60.8%, leading to an overestimated consumption reduction of sugar-sweetened soft drinks of up to 57.9%.

As exploring the role of heterogeneity in explaining household demand is important especially from marketing and policy perspectives,

☆ Generous financial support from USDA's Food Assistance and Nutrition Research Program (Cooperative Agreement 58-5000-5-0009) is gratefully appreciated. The views expressed here are those of the authors and cannot be attributed to the Economic Research Service or the U.S. Department of Agriculture. The authors wish to thank Mario Mazzochi, three anonymous reviewers, Chen Zhen, Yunhan Li, and audience members at the 2018 AAEA Meetings for helpful comments.

* Corresponding author.
E-mail address: jdorfman@uga.edu (J.H. Dorfman).

https://doi.org/10.1016/j.foodpol.2019.03.003
Received 9 March 2018; Received in revised form 18 January 2019; Accepted 13 March 2019
Available online 19 March 2019
0306-9192/ © 2019 Elsevier Ltd. All rights reserved.

and to extend earlier research in incorporating heterogeneity in demand estimation, this paper takes an innovative approach that combines a Panel Smooth Transition Regression (PSTR) model (Gonzalez et al., 2017) with an ECON transition function (Hood and Dorfman, 2015) to investigate the demand for regular carbonated sweetened beverages (CSBs) and beer. Specifically, our research seeks to determine the extent to which heterogeneous habit strength characterizes households' heterogeneous responses to price or income changes. At the final stage, an assessment of the potential effects of soda and beer taxes is implemented. Our policy simulation finds that households with more habit-based consumption are less sensitive to price or income changes. Since a higher habit level is significantly associated with higher consumption on CSBs and beer,[1] the overall effects of a sin tax would be highly over-estimated if heterogeneous consumer responses are not considered. Specifically, the forecast aggregate declines in soda and beer consumption due to a one-cent per ounce tax increase are found to be 20% and 26% larger, respectively, with homogeneous habit strength than with a model incorporating heterogeneity in habit strength. Because the top 10% of consumers in habit strength account for 18% and 23% of total consumption for CSBs and beer, respectively, it is important to account for their differential behavior in optimal policy design. Public health gains would come more from reductions in consumption from heavy users than occasional drinkers; thus, the tax rates may need to be raised further or alternative policies should be targeted at the more habitual and high-consumption consumers in order to meet public health goals.

In this paper, we make three major contributions to the empirical literature on demand analysis. First and foremost, this study contributes to the existing literature by introducing a new approach to relaxing the homogeneity assumption in demand estimation with habit effects. The proposed model allows continually varying household price/income elasticities as habit strength varies across households. The model can be easily extended to include other exogenous variables representing household heterogeneity and can also be applied to more sophisticated demand systems. Second, habit is generally explained as a repeated behavior pattern. The underlying hypothesis is that current consumption of one product is substantially affected by consumption of the same product in the previous period. Lagged purchases are usually included in demand models to demonstrate habits. An issue of this approach is that lagged response variables could suppress the explanatory power of other explanatory variables. Consequently, it may become difficult to identify the relationship between quantity and each explanatory variable independently because lagged purchase and other independent variables tend to change in unison. Moreover, the inference of price and income elasticities will be less precise, which weakens the statistical power of the model and the results of policy evaluation. This can be solved by including a smooth transition function of past purchases in the demand model, as we do here. Third, we introduce variance as a co-measurement of habit. Habitual consumption behavior refers to situations where a consumer's demand for a bundle of goods reveals very few differences across periods. Those loyal or even addicted consumers show different economic behavior than "normal" consumers and are more important in policy analysis due to their higher levels of consumption. A common practice in previous literature was to use only the sum of purchases in past periods, which does not take variation in past purchases into account when measuring habit. Occasional large purchases do not reflect a household buying for its own consumption the same as steady, consistent purchases. In previous models, both kinds of consumption might be classified equally as habitual behavior. Given this consideration, our variable of habit strength for a household is defined as the sum of consumption in the last eight quarters divided by the standard deviation of quarterly purchases. Thus, a large standard deviation due to inconsistent consumption behavior will lead to a lower habit strength. Incorporating variation in transition variables helps better measure customers' habitual behavior and allows us to better characterize individual policy responses, leading to better estimates of the aggregate response. Moreover, with each household's habit strength and demographics available, we are able to find the common characteristics of the habitual household thus helping with better targeted policies.

The outline of the paper is as follows. We lay out our research design and data description in Section 2 and 3, followed by the empirical demonstration in Section 4 that applies the heterogeneous habit model to both regular CSBs and beer in the United States, finding strong evidence for heterogeneity in both price and income elasticities across households. Finally, some policy implications and conclusions are provided in Section 5.

2. Empirical strategy

To examine the household's heterogeneous responses to price or income changes, this paper extends previous literature by proposing an ECON-PSTR model combined with simple demand functions to model habit strength that is heterogeneous across households.

2.1. Habit strength

Numerous studies have incorporated habit in demand models. It is well recognized among economists that demand in one period may depend on demand decisions in other periods (Becker and Murphy, 1988; Boyer, 1983). The literature has provided evidence of habit in not only addictive goods such as tobacco and alcohol but also non-addictive consumer goods (Arnade et al., 2008; Fuhrer, 2000; Heien and Durham, 1991; Holt and Goodwin, 1997). Particularly relevant here, Zhen et al. (2011) found the presence of habit in demand for nine categories of non-alcoholic beverages.

Different than the common practice in previous literature that habit is measured by purchases in past periods, we proposed a new index for habit strength, s_{ijt}, defined as:

$$s_{ijt} = \frac{(\sum_{d=1}^{8} Q_{ijt-d})}{\nu_{ij}} \tag{1}$$

where Q_{ijt} is the consumption of product j in quarter t for household i and ν_{ij} is the standard deviation of $(Q_{ijt-1}, Q_{ijt-2}, \cdots, Q_{ijt-8})$. Accordingly, habitual behavior is indexed by the sum of quarterly consumptions in the last two years over the standard deviation of those eight lagged values. Both quantity and variation are incorporated to depict past consumption behavior such that habit strength is strictly increasing in past consumption but is decreasing in consumption volatility. Given the same aggregate consumption in the past two years, a household with a more consistent purchasing pattern is considered to have greater habit strength in our model than a household with a volatile purchasing history. This formulation of habit strength is inspired by careful inspection of the data which reveals occasional spikes in household CSB consumption leading to large aggregate consumption that may be, in fact, caused by activities such as parties where CSBs were not necessarily consumed by the household alone. Including volatility of past consumption in our habit measure attempts to capture this feature of the data and more accurately measure true habitual consumption.

For each consumer, we calculate the mean habit strength over the sample period and plot histograms for CSBs and beer consumers, shown in Fig. 1 and Fig. 2. Both histograms show a highly right-skewed pattern, due to extremely high habit strength in a small minority of consumers with consistent and large purchases over the sample period.

[1] Pearson's correlation coefficient, ρ, between mean habit strength (see Eq. (1)) and mean consumption by household are calculated for CSBs and beer: ρ_{CSBs}=0.53(p-value < 0.0001); ρ_{beer}=0.51(p-value < 0.0001).

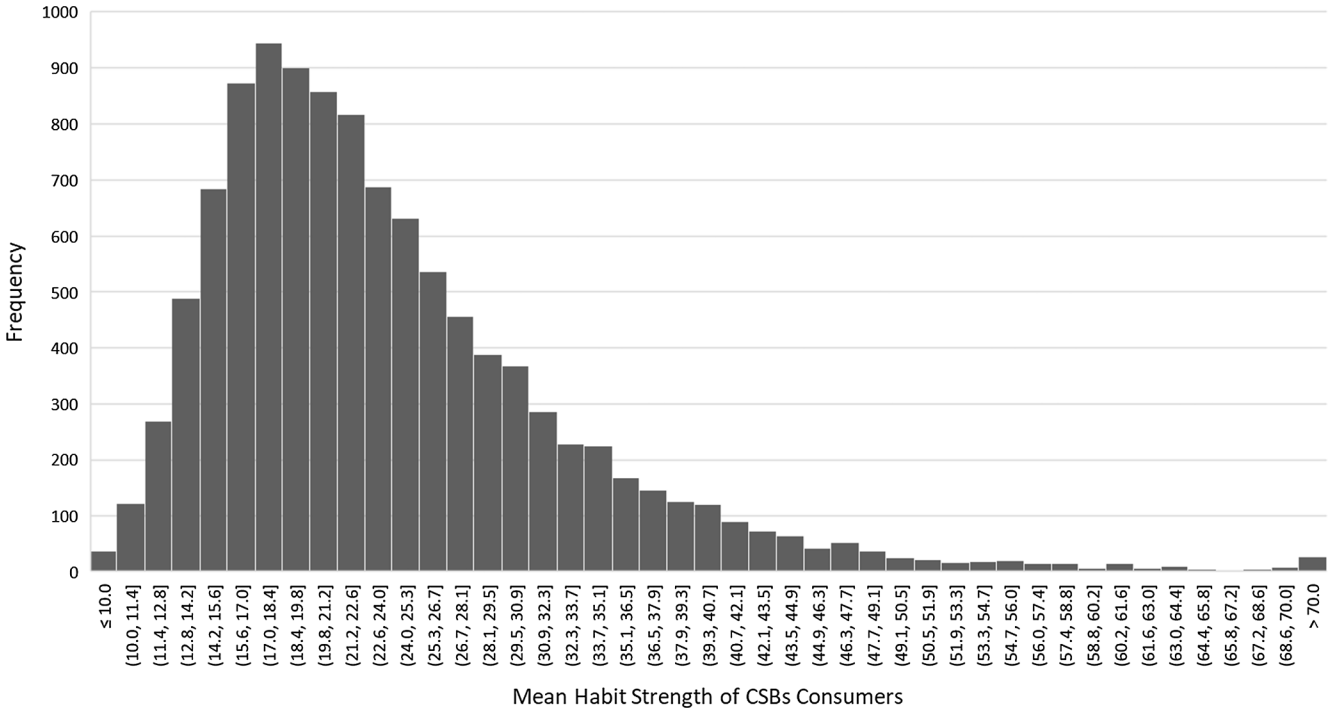

Fig. 1. Histogram of average habit strength over sample period – CSBs consumers.

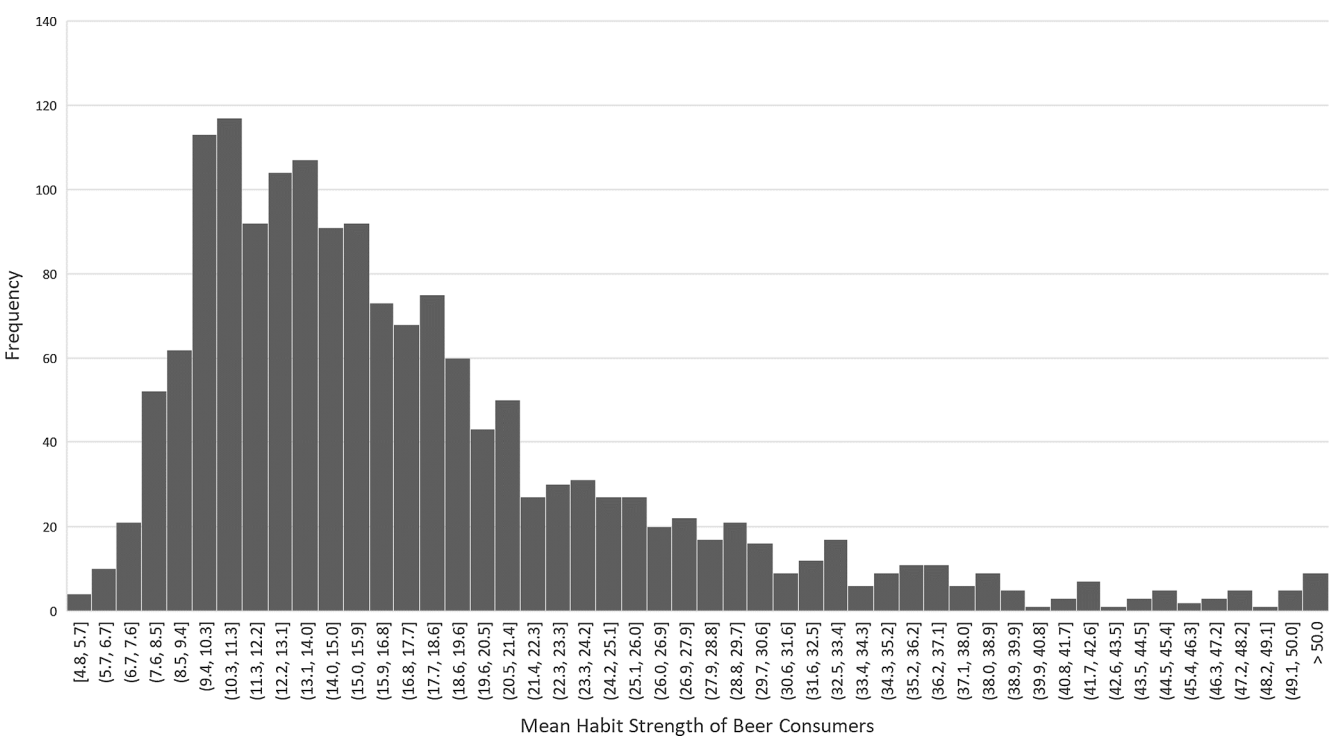

Fig. 2. Histogram of average habit strength over sample period – beer consumers.

2.2. ECON-PSTR model specification

Economists have increased their interest in the modeling of heterogeneity by such methods as latent class models and clustering analysis (Baker and Burnham, 2001; Bertail and Caillavet, 2008; Kikulwe et al., 2011; Ortega et al., 2011). These approaches can work well, but they reduce the modeling of heterogeneity down to a set of discrete values or groups. In contrast, we essentially create an infinite number of latent classes to represent the range of possible strengths of habit in beverage demand by using a demand model that nests a smooth transition function. Each household is treated differently based on the value of an index variable that summarizes the strength of habit in their household beverage purchasing decisions.

The Smooth Transition Regression (STR) model, initially developed by Bacon and Watts (1971), can be seen as a generalized regime switching model in such a way that the transition from one extreme regime to the other is not discrete, but smooth, with a function of the continuous transition variable. The PSTR model is a newly developed

type of STR model in the first version of Gonzalez et al. (2017) and Fok et al. (2005). Since its first appearance in 2005, the PSTR model has been applied to a wide variety of statistical and economic studies (e.g. Cheng and Wu, 2013; Delatte et al., 2012; Espinoza et al., 2011; Fok et al., 2005; Geng, 2011; Omay et al., 2018; Seleteng et al., 2013). In this paper, to estimate the impact of food tax policies on consumption, we use the PSTR approach to model continuous habit regimes, thus introducing heterogeneity in habit strength to demand studies. The PSTR model in this paper has a structure similar to Lundbergh et al. (2003), but is modified in a panel setting. The model is specified as follows:

$$lnQ_{ijt} = \mu_i + \lambda_t + f(x, \theta^{(1)})[1 - G(s_{ijt}, c_{jt}, \sigma_{jt}; \gamma)] + f(x, \theta^{(2)})G(s_{ijt}, c_{jt}, \sigma_{jt}; \gamma) + \varepsilon_{ijt} \tag{2}$$

where μ_i represent individual household fixed effects and λ_t represents quarter and year fixed effects, respectively, ε_{ijt} are the errors, lnQ_{ijt} is the natural logarithm of demand for product j at time t of household i, G is a smooth transition function, x is a vector of intercept and regressors including, prices, expenditures, and social demographics, θ is a vector of coefficients, and $f(x, \theta^{(k)})$ is the demand equation for regular CSBs or beer. Making things concrete, in the application that follows:

$$f(x, \theta^{(k)}) = x'\theta^{(k)} = \theta_1^{(k)} + \theta_2^{(k)} lnP_{jt} + \theta_3^{(k)} lnY_{it} + \theta_4^{(k)} Gender_i + \sum_{h=1}^{3} \theta_{5,h}^{(k)} Race_{hi} + \theta_6^{(k)} Edu_i + \theta_7^{(k)} Chd_{it},\ k = 1, 2 \tag{3}$$

where lnP_{jt} is the Törnqvist panel price index of product j at time t, $lnY_{i,t}$ is the natural logarithm of expenditure of household i at time t, $Gender_i$ is coded as 1 if the household head is male and zero otherwise, the $Race_{hi}$ are a set of binary indicators for the race of the household head ($h = 1$ for white, $h = 2$ for black, and $h = 3$ for Hispanic) with other races (Asian, American Indian, multiracial Americans, etc.) as the reference group. Edu_i is an indicator for the household head having at least a college degree, and $CHD_{i,t}$ is an indicator for household i having at least one child at time t.

The $f(x, \theta^{(1)})$ and $f(x, \theta^{(2)})$ are demand functions in two extreme regimes. Regime One represents the households who do not consume product j based on habit at all, implemented through the vector of parameters $\theta^{(1)}$, whereas, Regime Two represents households who have a strong habit of consistently consuming product j or are even addicted to it, expressed through a vector of parameters $\theta^{(2)}$. Most consumers behave consistent with some state in between the two extremes, with an infinite number of such regimes lying on that continuum and their location on the continuum expressed by the value of G. If consumers overall display little heterogeneity in the strength of their habits they will show similar responses to price or income changes; that is, the parameter vectors $\theta^{(1)}$ and $\theta^{(2)}$ will be similar. In this case, estimation results will be similar to those of a standard regression without a transition function.

Among numerous variations of the transition function G, the logistic STAR (LSTAR) model and the exponential STAR (ESTAR) model are the most commonly used. The transition function employed in this paper follows the ECON-STAR model (Hood and Dorfman, 2015), which is a variant of ESTAR model. In the ECON-STAR model, the transition function takes the form

$$G(s_t, v_t; \gamma, c, d) = 1 - exp\left\{-\gamma\left[\left(\frac{s_t - c}{\sigma_s}\right)\left(\frac{v_t - d}{\sigma_v}\right)\right]\right\} \tag{4}$$

where the speed-of-adjustment parameter $\gamma > 0$ is required, s_t and v_t are transition variables (such as rolling averages of past values of an economic indicator) in two adjacent regions, c and d are the minimum transition variable values over time for each location thus ensuring $(s_t - c)$ and $(v_t - d)$ are non-negative. Also, $(s_t - c)$ and $(v_t - d)$ are normalized by their standard deviations, σ_s and σ_v, respectively. With this specification, the value of the G(·) function will be zero at least once when $s_t = c$ or $v_t = d$.

In this paper, we generalize the transition function into a panel setting and offer modest improvements. The function $G(s_{ijt}, c_{jt}, \sigma_{jt}; \gamma)$ is a continuous and smooth function of s_{ijt}, $c_{jt}$and σ_{jt}, taking the form:

$$G(s_{ijt}, c_{jt}, \sigma_{jt}; \gamma) = 1 - exp\left\{-\gamma\left[\Phi\left(\frac{s_{ijt} - c_{jt}}{\sigma_{jt}}\right)\right]\right\} \tag{5}$$

where $\gamma > 0$ is required, s_{ijt} is the habit strength as defined in Eq. (1), c_{jt} is the mean value of s_{ijt} across households, and σ_{jt} is the standard deviation of s_{ijt}, $s_{ijt} - c_{jt}$ is normalized by σ_{jt} to make the speed-of-adjustment parameter γ unit free, and $\Phi(\cdot)$ is the cumulative density function (cdf) of the standard normal distribution. Embedding the term $\frac{s_{ijt} - c_{jt}}{\sigma_{jt}}$ in the cdf of the standard normal distribution gives an almost linear shape for values of s_{ijt} within 2 standard deviations about zero but has flatter tails as the value of s_{ijt} goes to extremes when compared with the linear term $\frac{s_t - c}{\sigma_s}$ in Eq. (4). This property enables coefficient estimates to be less influenced by extreme values of s_{ijt}, thus yielding more robust coefficient estimates. In addition, $\Phi\left(\frac{s_{ijt} - c_{jt}}{\sigma_{jt}}\right)$ also satisfies the positivity requirement thus keeping the value of transition function bounded between 0 and 1 when combined with the imposed restriction of $\gamma > 0$.

2.3. The estimation of a PSTR model

The estimation of the PSTR model is accomplished by a two-step procedure (Gonzalez et al., 2017). The first step is eliminating fixed effects (μ_i and λ_t) by subtracting the individual- and time-specific means on both sides of the equation. The second step is to estimate the model with the transformed data by nonlinear least squares (NLS). To simplify algebra and without loss of generality, we define a PSTR demand model with only one product for household i at time t:

$$lnQ_{it} = \mu_i + \lambda_t + x_{it}'\theta^{(1)}G(z_{it}; \gamma) + x_{it}'\theta^{(2)}[1 - G(z_{it}; \gamma)] + \varepsilon_{it} \tag{6}$$

where $z_{it} = \frac{s_{it} - c_t}{\sigma_t}$. First, we subtracted the mean for each household over time and the mean for each time period across all households from the dependent variable and the residuals to construct centered variables, $Q_{it}{}^c$ and $\varepsilon_{it}{}^c$:

$$Q_{it}{}^c = lnQ_{it} - ln\bar{Q}_i - ln\bar{Q}_t \tag{7}$$

$$\varepsilon_{it}{}^c = \varepsilon_{it} - \bar{\varepsilon}_i - \bar{\varepsilon}_t \tag{8}$$

where $ln\bar{Q}_i = \frac{1}{T_i}\sum_{t=1}^{T_i} lnQ_{it}$, $ln\bar{Q}_t = \frac{1}{N_t}\sum_{i=1}^{N_t} lnQ_{it}$, $\bar{\varepsilon}_i = \frac{1}{T_i}\sum_{t=1}^{T_i} \varepsilon_{it}$, $\bar{\varepsilon}_t = \frac{1}{N_t}\sum_{i=1}^{N_t} \varepsilon_{it}$, T_i is the number of time periods that household i was observed in the panel, and N_t is the number of households observed at time t. For the terms $x_{it}'G(z_{it}; \gamma)$ and $x_{it}'[1 - G(z_{it}; \gamma)]$, we make the following transformations:

$$A_{it}{}^c(z_{it}; \gamma) = x_{it}'G(z_{it}; \gamma) - \bar{A}_i' - \bar{A}_t' \tag{9}$$

$$B_{it}{}^c(z_{it}; \gamma) = x_{it}'[1 - G(z_{it}; \gamma)] - \bar{B}_i' - \bar{B}_t' \tag{10}$$

where $\bar{A}_i = \frac{1}{T_i}\sum_{t=1}^{T_i} x_{it}G(z_{it}; \gamma)$, $\bar{A}_t = \frac{1}{N_t}\sum_{i=1}^{N_t} x_{it}G(z_{it}; \gamma)$, $\bar{B}_i = \frac{1}{T_i}\sum_{t=1}^{T_i} x_{it}[1 - G(z_{it}; \gamma)]$, and $\bar{B}_t = \frac{1}{N_t}\sum_{i=1}^{N_t} x_{it}[1 - G(z_{it}; \gamma)]$. Therefore, each row of the new centered design matrix becomes $x_{it}{}^c(z_{it}; \gamma)' = [A_{it}{}^c(z_{it}; \gamma)'; B_{it}{}^c(z_{it}; \gamma)']$.

After eliminating fixed effects by data transformation, we can apply nonlinear least squares (NLS) to estimate the coefficients that minimize the concentrated sum of squared errors:

$$SSE^c(\gamma) = \sum_{i=1}^{N}\sum_{t=1}^{T}[Q_{it}{}^c - x_{it}{}^c(z_{it}; \gamma)'\hat{\theta}(\gamma)]^2 \tag{11}$$

where $\hat{\theta}(\gamma)$ is obtained by ordinary least squares (OLS) at each iteration in the non-linear optimization and is strongly dependent on the quality

of initial values. A grid search for the parameter γ in the transition function $G(z_{it}; \gamma)$ is applied to obtain the starting value of γ.[2]

3. Data

3.1. Data description

Our primary source of data is the Nielsen Homescan household panel for the years 2008 to 2015. More than 100,000 households across the U.S. record information on shopping trips and purchased items using an optical scanner on a weekly basis over a period of at least a year. Every recorded transaction contains information including the Universal Product Code (UPC), quantity, price paid, size, single or multipack, and brand. The Nielsen Homescan data is available to academic researchers through a partnership between Nielsen and USDA-Economic Research Service (ERS). The biggest advantage of the Homescan panel data set is that the sample is nationally representative. The participating households reside in fifty-two Nielsen markets and nine remaining areas in the United States. Household survey weights provided by Nielsen can be used to create national estimates of household purchases. Further, the Homescan dataset includes almost all U.S. retailers including mass merchants such as Walmart.

To analyze heterogeneous demand responsiveness, we focus on CSBs and beer products. CSBs is one of the four categories included in Sugary-sweetened beverages (SSBs), which is a common target of soda tax policies. The other three categories are regular non-diet fruit juices, non-diet sports and energy drinks, and all other SSBs. Among the four categories of SSBs, regular CSBs are the most prevalent SSB type across all years for all age groups except children (for whom fruit drinks were the most prevalent in 1999 and 2005) (Han and Powell, 2013). We choose only CSBs as our research subject since it is the most representative group and analyzing the effect of soda tax on the full categories of SSBs would generally involve product aggregation or estimation of a system of demand equations, which is out of the research scope of this paper.

We apply the following screens to ensure the data used consist only of households who consistently recorded their purchasing sometime between 2008 and 2015. Households who fail to satisfy any of the following criteria are excluded from the dataset:

(i) Each household must be on the panel for at least nine consecutive quarters since we track household habits based on two years of purchasing history.
(ii) Each household must have at least one shopping trip per quarter when on the panel. A quarter is considered long enough to identify that the household has stopped recording its purchases.
(iii) The consumption of regular CSBs or beer by the household must be positive for at least one quarter because the target population in our study is CSBs or beer consumers.

We also deleted observations with abnormally large or small prices. Specifically, we deleted transactions in which unit price for a specific product was more than five times or less than one-fifth of the sample mean price.

3.2. Summary statistics

Table 1 summarizes the demographics of the households of different habit strength levels. For each product, CSBs and beer, the households are classified into four groups, with Group 1 representing the quartile of

[2] The Model and NLIN procedures in SAS 9.4 are used for model estimation. For future researchers who are interested in applying PSTR models, an R package (PSTR) written by Y. Yang provides a useful tool for model setup, estimation, and evaluation.

Table 1
Homescan sample summary statistics, by products and habit strength quartiles.

	Group	Range of Habit Strength[a]	Number of Households	Mean Habit Strength	Mean Household Size	Mean Household Income[b]	Percentage of Male Household Head	Race of Household Head				At least one Child	Household Head holding at least College Degree
								White	Black	Hispanic	Others		
CSBs Households (N = 10921)	1	$>Q_3$	2731	36.27	2.38	8.29	35.77%	92.27%	4.25%	3.22%	0.26%	22.38%	38.04%
				(9.08)	(1.11)	(2.84)	(9.17e−3)	(5.11e−3)	(3.86e−3)	(3.38e−3)	(9.68e−4)	(7.98e−3)	(3.86e−4)
	2	$(Q_2, Q_3]$	2730	24.40	2.37	8.36	30.11%	89.12%	6.23%	3.85%	0.81%	19.67%	40.55%
				(1.74)	(1.15)	(2.94)	(8.78e−3)	(5.96e−3)	(4.62e−3)	(3.68e−3)	(1.71e−3)	(7.61e−3)	(9.40e−3)
	3	$(Q_1, Q_2]$	2730	19.43	2.42	8.43	31.14%	88.10%	7.18%	4.18%	0.55%	17.47%	39.93%
				(1.23)	(1.20)	(2.87)	(8.86e−3)	(6.20e−3)	(4.94e−3)	(3.83e−3)	(1.41e−3)	(7.27e−3)	(9.37e−3)
	4	$\leq Q_1$	2730	14.79	2.51	8.44	29.08%	84.84%	9.19%	5.13%	0.84%	15.16%	41.43%
				(1.85)	(1.30)	(2.97)	(8.69e−3)	(6.86e−3)	(5.53e−3)	(4.22e−3)	(1.75e−3)	(6.86e−3)	(9.43e−3)
Beer Households (N = 1613)	1	$>Q_3$	404	30.19	2.03	8.27	44.42%	90.07%	5.46%	4.22%	0.25%	8.93%	33.75%
				(9.90)	(0.91)	(2.91)	(2.47e−2)	(1.49e−2)	(1.13e−2)	(1.00e−2)	(2.48e−3)	(1.42e−2)	(2.35e−2)
	2	$(Q_2, Q_3]$	403	17.80	2.14	8.78	41.69%	87.10%	8.68%	3.72%	0.50%	10.17%	35.48%
				(1.58)	(0.92)	(2.51)	(2.46e−2)	(1.67e−2)	(1.40e−2)	(9.43e−3)	(3.50e−3)	(1.51e−2)	(2.38e−2)
	3	$(Q_1, Q_2]$	403	13.40	2.22	9.02	42.43%	88.83%	6.20%	3.97%	0.99%	12.90%	42.93%
				(1.08)	(1.03)	(2.69)	(2.46e−2)	(1.57e−2)	(1.20e−2)	(9.73e−3)	(4.94e−3)	(1.67e−2)	(2.47e−2)
	4	$\leq Q_1$	403	9.62	2.23	9.13	40.94%	84.37%	11.17%	3.72%	0.74%	14.39%	47.15%
				(1.34)	(0.95)	(2.67)	(2.45e−2)	(1.81e−2)	(1.57e−2)	(9.43e−3)	(4.28e−3)	(1.75e−2)	(2.49e−2)

[a] Q_1, Q_2, Q_3 represent the first, second, and third quartile of the habit strength, respectively.
[b] Household Income is a categorical variable where annual income ranges from \$0 to \$9999 is coded as 1, \$10,000 to \$11,999 as 2, \$12,000 to \$14,999 as 3, \$15,000 to \$19,999 as 4, \$20,000 to \$24,999 as 5, \$25,000 to \$34,999 as 6, \$35,000 to \$44,999 as 7, \$45,000 to \$49,999 as 8, \$50,000 to \$59,999 as 9, \$60,000 to \$69,999 as 10, \$70,000 to \$99,999 as 11, and \$100,000 and greater as 12.

Table 2
Parameter estimates for regular CSBs and beer.

Parameter Estimates for Regular Carbonated Sweetened Beverages and Beer					
Scheme 1		Regular Carbonated Sweetened Beverages		Beer	
	Model	Linear Demand Model	PSTR Model	Linear Demand Model	PSTR Model
	Intercept	4.284*	3.002*	1.959*	0.002
		(0.061)	(1.012)	(0.531)	(0.003)
	Natural log *of Price*	**−1.055***	**−1.598***	**−1.089***	**−1.936***
		(0.082)	(0.197)	(0.128)	(0.154)
	Natural log *of Expenditure*	**0.347***	**0.261***	**0.677***	**0.620***
		(0.006)	(0.010)	(0.003)	(0.005)
	Gender of Household	0.055*	0.058*	0.198*	0.164*
		(0.013)	(0.020)	(0.001)	(0.006)
	Race1	0.103*	0.141*	0.213*	0.249*
		(0.037)	(0.053)	(0.001)	(0.013)
	Race2	−0.126*	−0.037*	−0.171*	−0.143*
		(0.043)	(0.062)	(0.001)	(0.018)
	Race3	−0.019	0.127	0.015*	0.194*
		(0.061)	(0.097)	(0.002)	(0.022)
	Education Level of Household Head	−0.096*	−0.045*	−0.266*	−0.253*
		(0.013)	(0.019)	(0.000)	(0.006)
	Two or more children in the household	0.121*	0.015	−0.220*	−0.233*
		(0.016)	(0.022)	(0.001)	(0.007)
Gamma	*Gamma*		**1.385***		**1.735***
			(0.044)		(0.036)
Scheme 2	*Intercept*		0.823*		7.301*
			(0.210)		(0.667)
	Natural log *of Price*		**−0.126***		**−0.191***
			(0.023)		(0.002)
	Natural log *of Expenditure*		**0.384***		**0.689***
			(0.015)		(0.138)
	Gender of Household		0.015		1.388*
			(0.028)		(0.227)
	Race1		−0.094		−1.705
			(0.076)		(1.335)
	Race2		−0.224*		−1.309
			(0.090)		(0.766)
	Race3		−0.254		−7.728*
			(0.132)		(0.951)
	Education Level of Household Head		−0.154*		−0.544*
			(0.027)		(0.231)
	Two or more children in the household		0.294*		−0.014
			(0.032)		(0.326)
	N	243,859	243,859	40,291	40,291
	SSE	181,180	156,250	57,810	52,330
	Number of Parameters	9	19	9	19
	AIC	−31446.77	−47104.58	6335.30	4612.66
	BIC	−31416.28	−47040.23	6358.75	4662.16

Note: Heteroscedasticity-consistent standard error in parenthesis.
* Significant at 5% significance level.

households with the greatest habit strength, that is, the most habitual households, Group 2 representing the second most habitual households, and so on.

For CSBs, the most habitual consumers are associated with a smaller household size, a lower annual household income, a higher probability of having a male and white household head without college degree, and a higher probability of having one or more children, on average, than other consumers. We do not find a significant difference in mean household size between any two of Group 1, 2, and 3. However, mean household size of Group 4 is significantly larger than each of Group 1 to 3, although the largest difference is only 0.14, indicating the least habitual households are associated with a slightly larger household size on average. Similarly, significant differences in mean income are found only between Group 1 and 3, and between Group 1 and 4, with the most habitual households associated with the lowest mean income in the sample. The most habitual households are more likely to have a male household head than any of the other groups, and the least habitual households are less likely to have a male household head than any of the other groups. A more habitual household is associated with a higher probability of having a white household head, hence a lower probability of having a non-white household head and is also associated with a higher probability of having at least one child. Last but not least, the heads of the most habitual households are found to be least likely to hold a college degree or higher while the heads of the least habitual households are the most likely to have a college or higher degree.

Similarly, for beer, the most habitual consumers are associated with a smaller household size, lower annual household income, a higher probability of having a male and white household head without college degree, and a lower probability of having one or more children, on average, than the other consumers. Specifically, statistically significant differences in mean household size are found between Group 1 and each of Group 2 to 4, although the largest difference is only 0.2, indicating the most habitual households are associated with a slightly smaller household size on average. The mean annual household income of Group 1 is found to be the lowest among the four groups. As with CSBs consumers, the most habitual households are found to be the most likely to have a male household and the least habitual households are

the least likely to have a male household head among all the groups. Also, Group 1 is associated with the highest probability of having a white household head, and Group 4 with the least. Moreover, more habitual households are found to have a lower probability of having at least one child, and a lower probability of holding a college degree or higher.

3.3. Price index

To reduce the unit value bias (Cox and Wohlgenant, 1986; Deaton, 1988), we created a quarterly superlative Törnqvist price index for product j for each state. Let an entity be a unique combination of location and time. For example, the same location (e.g., a county) in period t and period $t+1$ is considered as two distinct entities in index formulas. The Törnqvist price index is defined as

$$p_T^{0k} = exp\left\{0.5 \sum_{v \in 0k} (s_v^0 + s_v^k)\ln(p_v^k/p_v^0)\right\} \tag{12}$$

where p_v^k is the is the price of product v in entity k; p_v^0 and q_v^0 are the base price and quantity of product v, respectively; v_{0k} denotes the common set of items sold in both base 0 and entity k, and s_v^0 and s_v^k are budget shares of product v in base 0 and entity k.

4. Empirical results

This section presents demand estimation for regular CSBs and beer, respectively, with both homogeneous and heterogeneous habit strength, followed by a simulation of demand change when soda and beer taxes are imposed. We assess a sin tax effect because taxes on energy-dense, low-nutrient foods such as sugar-sweetened beverages (SSBs) is the most commonly proposed anti-obesity policy (Cawley, 2015). From the policy maker's perspective, the stated goal of the soda tax is to reduce consumption of these beverages, stem the obesity epidemic, fund health-related initiatives, and raise much-needed revenue to offset a large state budget deficit (New York State Department of Health, 2010).

4.1. Carbonated sweetened beverages

First, the linear demand model in Eq. (3) is estimated for regular CSBs. As shown in Table 2, the coefficient on lnP_{jt}, β_1, is estimated to be -1.055 and statistically significant, indicating that a one percent increase in prices would result in a 1.055% decrease in quantity consumed.

Next, the ECON-PSTR model of heterogeneous habit strength is estimated. The fourth column of Table 2 presents the PSTR model results, which are further illustrated in Fig. 4. In Fig. 4, the x-axis represents the value of G, which is bounded between 0 (no habit at all) and 1 (addiction). Based on our estimation, the value of G ranges across households from 0.07 to 0.99. The solid line shows a household's own-price elasticity decreases in magnitude from -1.422 to -0.410 as a household's habit strength increases from 0.07 to 0.99. Thus, for the more habitual consumers, who are the primary target of public health policy makers, sin tax style policies are not as effective as for those who are not as habitual in consumption. The dotted line represents the expenditure elasticity of each household under different habits. The expenditure elasticity increases from 0.276 to 0.365 as a household's habit level increase from 0.07 to 0.99. These findings are in line with expected behavior. Habitual consumers would allocate more spending on goods they are most habituated to as their total expenditure increases. Additionally, model selection criteria (AIC and BIC) are smaller when the PSTR model is applied, providing evidence in favor of the heterogeneous habit model for these data. Fig. 3 plots the value of G functions against mean habit strength of CSBs and beer consumers.

To understand the influence of a soda tax on consumption, we simulate a scenario where a one-cent per ounce tax[3] is levied on regular CSBs. We assume that the tax is fully passed through to retail prices. Applying the parameters of the homogeneous habit model, the estimated decrease in purchases of each household is illustrated in Fig. 5, where blue bars and orange bars indicate demand drop under homogeneous and heterogeneous habit strength assumptions, respectively. With a model of homogeneous habit, the percentage change is the same across households given equal price elasticities. Consequently, the absolute decrease in purchases of habitual households is larger than for non-habitual households when facing the same percentage increase in prices. Based on the reactions simulated for each household with homogeneous habit, the one-cent per ounce soda tax would result in an aggregate 28.84% decrease in household CSBs.

From the results of the heterogeneous habit strength model, the own-price elasticity ranges from -1.422 to -0.410. As the orange bar in Fig. 5 indicates, the net decrease in purchases of habitual households is not much larger than that of non-habitual households because the households with the largest purchases tend to be more price inelastic. Based on this simulation, a one-cent per ounce soda tax would result in a 24.11% decrease in overall consumption. Thus, the demand response to a soda tax when heterogeneous habit strength is allowed for in the model is considerably less than under the standard, homogeneous habit model. In fact, when individual heterogeneity is ignored, the overall decrease in household regular CSBs demand would be overestimated by 4.73 percentage points.

4.2. Beer

To further demonstrate the importance of allowing heterogeneity in habit, we perform a second simulation, this time applied to beer purchasing behavior. Similar results will suggest that the specific product chosen has not biased our results. Intuitively, beer is a more habit-forming product than regular CSBs. As before, first, a constant habit regression for beer demand is conducted using Eq. (3). The coefficient on lnP_{jt} is estimated to be -1.089 and statistically significant, indicating that a one percent increase in prices would result in a 1.089% decrease in average consumption of beer. Linear regression results are presented in the fifth column of Table 2.

Next, we estimate our heterogeneous habit PSTR model for beer. Column 6 in Table 2 presents the nonlinear regression results, and Fig. 6 shows the relationship between habit, price elasticity, and income elasticity. Similar to Fig. 4, the x-axis measures the level of habit, which ranges from 0.21 to 0.99, suggesting that even the least habitual consumers in the dataset have a moderate level of habit in purchasing beer. The absolute own-price elasticity decreases from 1.323 to 0.357 and the expenditure elasticity increases slightly from 0.643 to 0.679 as a household's habit level increases. Additionally, model selection statistics (AIC and BIC) are again smaller when the PSTR model is applied, indicating that incorporating heterogeneous habit is the preferred model specification. The shape of G functions with respect to mean household habit strength of beer consumers is illustrated in Fig. 3.

We again simulate a scenario of an additional one-cent per ounce tax being implemented and fully passed through to retail prices. Based on the results of the homogeneous habit model, with its price elasticity of -1.089, the tax would result in a 15.76% decrease in overall beer sales. The decrease in purchases among consumers with different habit strengths when habit strength is modeled as constant is illustrated by the blue bar in Fig. 7. Under our heterogeneous habit model, a one-cent per ounce beer tax would result in only a 12.49% decrease in overall

[3] Some past literature used half-cent per ounce soda tax in policy simulations (Dharmasena and Capps, 2012; Lin et al., 2011; Zhen et al., 2014, 2011). In the November 4, 2014 election, Berkeley, California, enacted the first soda tax in the United States, establishing one-cent per ounce tax on sugary drinks. To make the simulations more relevant to current public policy, we use one-cent per ounce as the proposed soda tax.

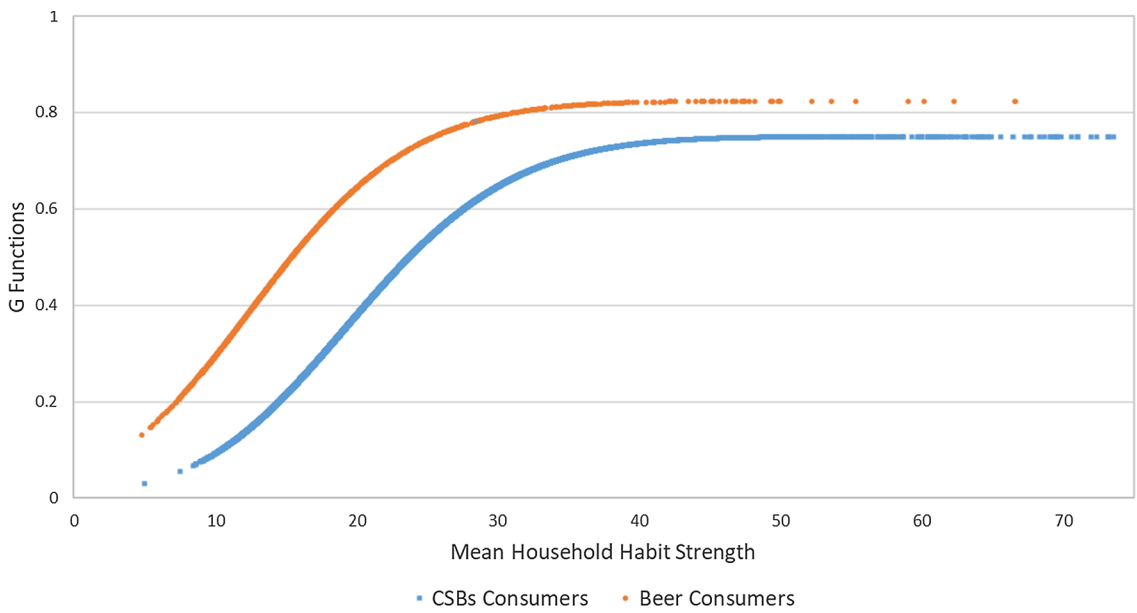

Fig. 3. The value of G – functions against average habit strength over sample period.

sales of beer. The reduced purchases among consumers at different habit levels when heterogeneous habit strength is modeled are illustrated by the orange bar in Fig. 7. The figure makes clear that beer is a more habitual product, and the more habitual a household is, the less sensitive to price changes they are. Like the results from regular CSBs, the net decrease from more habitual households is quite similar to less habitual households. If individual heterogeneity is ignored, the overall decrease in household beer demand would be overestimated by 3.27 percentage points. In other words, with the stronger habit of beer demand across the spectrum, incorporating heterogeneous habit leads to estimates of responsiveness to a beer tax only 80% of that suggested by the uniform habit model.

5. Conclusions and policy implications

Previous food demand research typically has found an own-price elasticity of −1.05 to −1.1 for the category of regular CSBs (Bergtold et al., 2004; Heien and Wessells, 1990; Pittman, 2005). An estimated own-price elasticity of −1.2 was applied by Yale University's Rudd Center to calculate the tax revenues generated by a soda tax (Rudd Center for Food Policy and Obesity (2010)). More recently, Zhen et al. (2014) estimated that the price elasticity of regular CSBs as −1.035 based on the Nielsen Homescan Dataset (also used here). With previous published estimates of price elasticities, the percentage drop in aggregated purchase of regular CSBs due to a one-cent per ounce soda tax is between the interval of 28.71% to 32.81% given our calculated CSBs prices, which contains our simulated results (28.84%) in the homogeneous habit model. Compared with the more general and realistic heterogeneous habit strength model, assuming constant habit strength leads to significant overestimation of the effect of a soda tax on CSBs purchases.

In this paper, we proposed an innovative procedure to investigate the demand for regular CSBs and beer under heterogeneous habit

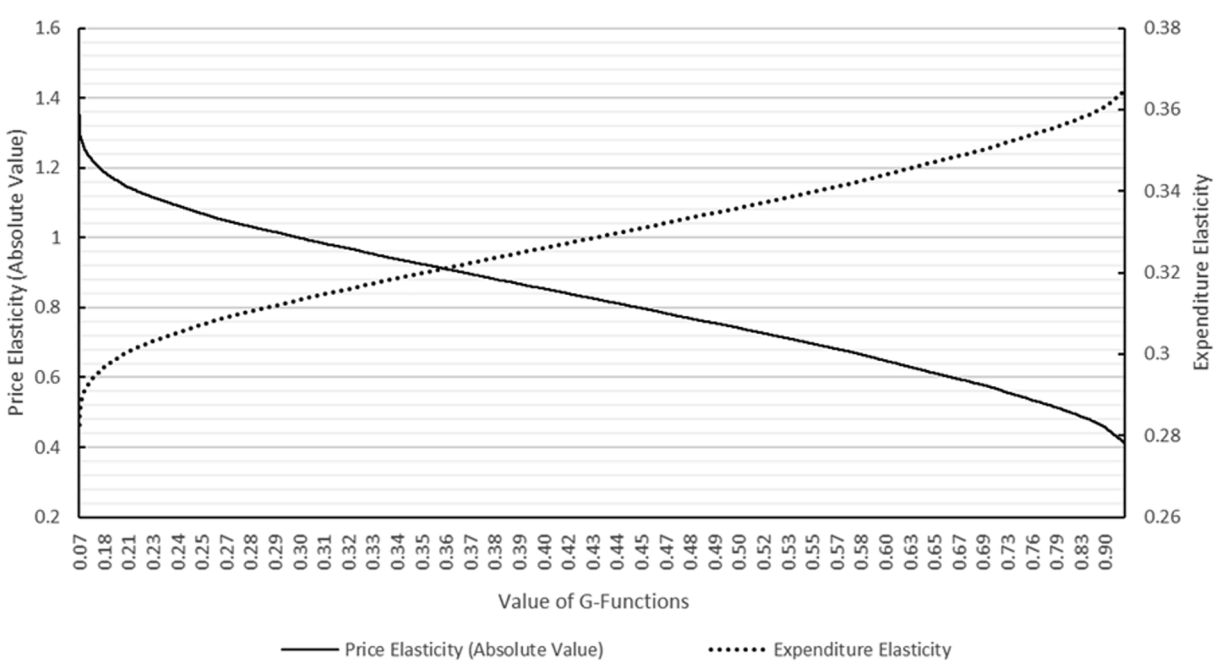

Fig. 4. Habit, price elasticity and expenditure elasticity for regular CSBs.

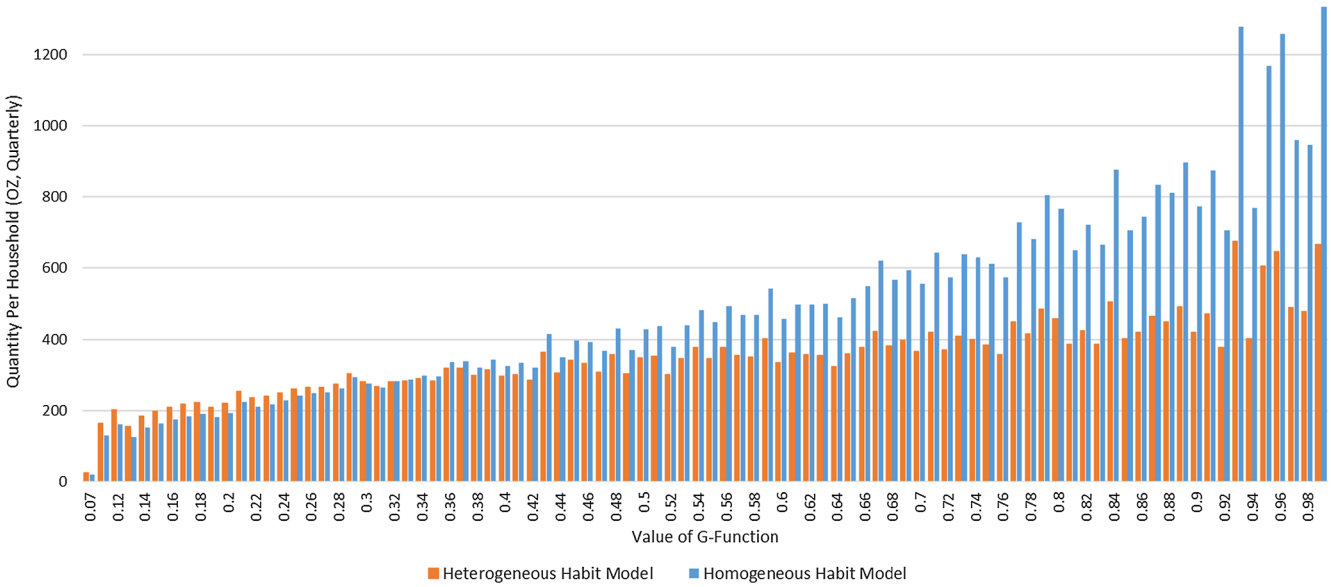

Fig. 5. Decrease in demand of regular CSBs due to soda tax under both heterogeneous and homogeneous habit assumptions.

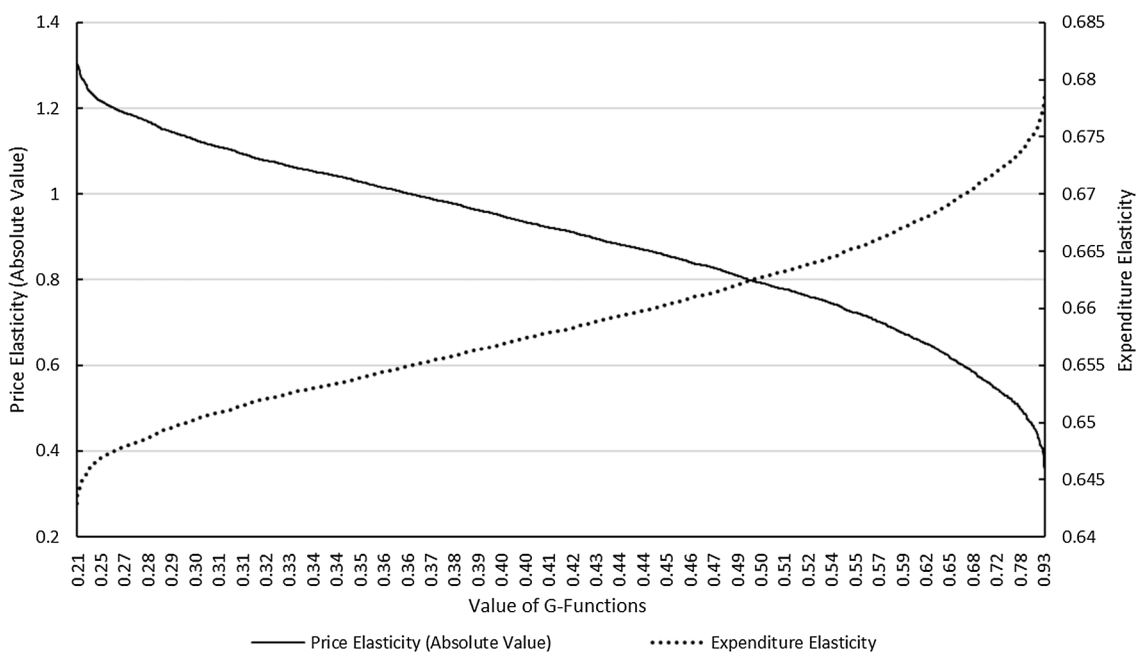

Fig. 6. Habit, price elasticity and expenditure elasticity for beer.

strength. To demonstrate how important this model generalization is for policy analysis, we reported an analysis of the potential effects of a one-cent per ounce soda tax and a one-cent per ounce beer tax. Our results shed light on the importance of incorporating individual heterogeneity when conducting research on public food policies. Habitual consumers have the strongest preference for a particular product and are less price sensitive than those with weaker habit effects. Because those with stronger habit effects have more inelastic demands and consume larger than average quantities, ignoring the heterogeneity of habit leads to an aggregation bias that could lead to faulty policy analyses. Consequently, the welfare measurement of the policy may be inaccurate due to price and expenditure insensitivity of highly habitual consumers. Table 1 provides possible criteria related to demographics to screen the highly habitual consumers at whom a more specific policy could target.

The ECON-PSTR model of habit strength introduced here reveals heterogeneous consumption patterns for regular CSBs and beer, leading to different responses to public policy among people. If individual heterogeneity in habit strength is ignored, the overall effect of soda tax and beer tax would be overestimated by 4.73 and 3.27 percentage points respectively. In other words, the aggregate declines in soda and beer consumption are over-estimated by 20% and 26%, respectively, under the homogeneity assumption. This implies sin and food taxes will be less successful at discouraging consumption than predicted by constant-habit demand models. Our analysis also suggests food policies should be designed with consideration of heterogeneity in consumers' habits and their varying sensitivity to price and income changes. In particular, public health professionals and nutrition experts don't simply want to reduce consumption of sugar sweetened beverages and beer, they specifically want to reduce consumption among the heaviest consumers of these unhealthy beverages. Once heterogeneity of habit strength is accounted for, we find that sin taxes have less effect on

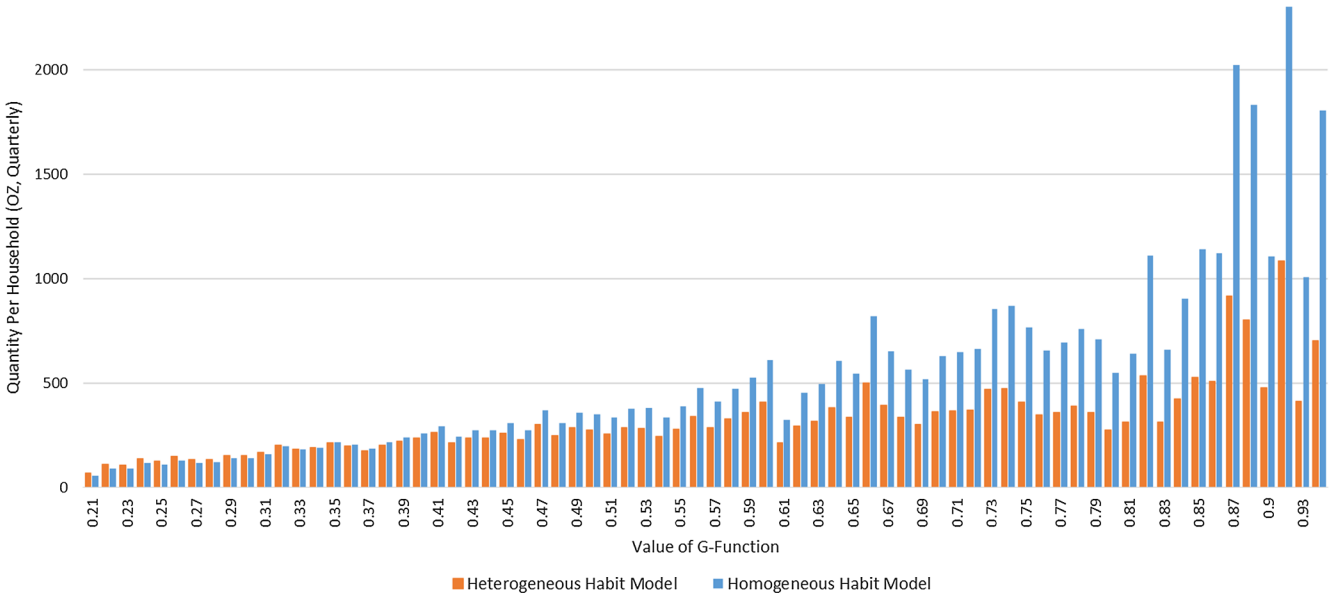

Fig. 7. Decrease in demand of beer due to beer tax under both heterogeneous and homogeneous habit assumptions.

exactly these targeted consumers. Thus, our results suggest that from a public health perspective, policies beyond simple price adjustments are needed to achieve the desired objectives.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodpol.2019.03.003.

References

Andreyeva, T., Long, M.W., Brownell, K.D., 2010. The impact of food prices on consumption: a systematic review of research on the price elasticity of demand for food. Am. J. Public Health 100 (2), 216–222.

Arnade, C., Gopinath, M., Pick, D., 2008. Brand inertia in US household cheese consumption. Am. J. Agric. Econ. 90 (3), 813–826.

Bacon, D.W., Watts, D.G., 1971. Estimating the transition between two intersecting straight lines. Biometrika 58 (3), 525–534.

Baker, G.A., Burnham, T.A., 2001. Consumer response to genetically modified foods: market segment analysis and implications for producers and policy makers. J. Agric. Resour. Econ. 26, 387–403.

Becker, G.S., Murphy, K.M., 1988. A theory of rational addiction. J. Polit. Econ. 96 (4), 675–700.

Bergtold, J., Akobundu, E., Peterson, E.B., 2004. The FAST method: estimating unconditional demand elasticities for processed foods in the presence of fixed effects. J. Agric. Resour. Econ. 29, 276–295.

Bertail, P., Caillavet, F., 2008. Fruit and vegetable consumption patterns: a segmentation approach. Am. J. Agric. Econ. 90 (3), 827–842.

Boyer, M., 1983. Rational demand and expenditures patterns under habit formation. J. Econ. Theor. 31 (1), 27–53.

Cawley, J., 2015. An economy of scales: a selective review of obesity's economic causes, consequences, and solutions. J. Health Econ. 43, 244–268.

Cheng, C., Wu, P., 2013. Nonlinear earnings persistence. Int. Rev. Econ. Financ. 25, 156–168.

Cox, T.L., Wohlgenant, M.K., 1986. Prices and quality effects in cross-sectional demand analysis. Am. J. Agric. Econ. 68 (4), 908–919.

Deaton, A., 1988. Quality, quantity, and spatial variation of price. Am. Econ. Rev. 418–430.

Delatte, A., Gex, M., López-Villavicencio, A., 2012. Has the CDS market influenced the borrowing cost of european countries during the sovereign crisis? J. Int. Money Financ. 31 (3), 481–497.

Dharmasena, S., Capps Jr, O., 2012. Intended and unintended consequences of a proposed national tax on sugar-sweetened beverages to combat the US obesity problem. Health Econ. 21 (6), 669–694.

Duffy, M., 2003. Advertising and food, drink and tobacco consumption in the united kingdom: a dynamic demand system. Agric. Econ. 28 (1), 51–70.

Espinoza, R., Leon, H., Prasad, A., 2011. When should we worry about inflation? World Bank Econ. Rev. 26 (1), 100–127.

Fok, D., Van Dijk, D., Franses, P.H., 2005. A multi-level panel STAR model for US manufacturing sectors. J. Appl. Economet. 20 (6), 811–827.

Fuhrer, J.C., 2000. Habit formation in consumption and its implications for monetary-policy models. Am. Econ. Rev. 90 (3), 367–390.

Geng, N., 2011. The dynamics of market structure and firm-level adjustment to india's pro-market economic liberalizing reforms, 1988–2006: a time varying panel smooth transition regression (TV-PSTR) approach. Int. Rev. Econ. Financ. 20 (4), 506–519.

Gonzalez, A., Teräsvirta, T., Van Dijk, D., Yang, Y., 2017. Panel Smooth Transition Regression Models. University of Uppsala Working Paper.

Goryakin, Y., Monsivais, P., Suhrcke, M., 2017. Soft drink prices, sales, body mass index and diabetes: evidence from a panel of low-, middle-and high-income countries. Food Policy 73, 88–94.

Haden, K., 1990. The demand for cigarettes in Japan. Am. J. Agric. Econ. 72 (2), 446–450.

Han, E., Powell, L.M., 2013. Consumption patterns of sugar-sweetened beverages in the united states. J. Acad. Nutr. Diet. 113 (1), 43–53.

Härkänen, T., Kotakorpi, K., Pietinen, P., Pirttilä, J., Reinivuo, H., Suoniemi, I., 2014. The welfare effects of health-based food tax policy. Food Policy 49, 196–206.

Heien, D., Durham, C., 1991. A test of the habit formation hypothesis using household data. Rev. Econ. Stat. 73, 189–199.

Heien, D., Wessells, C.R., 1990. Demand systems estimation with microdata: a censored regression approach. J. Bus. Econ. Stat. 8 (3), 365–371.

Herzfeld, T., Huffman, S., Rizov, M., 2014. The dynamics of food, alcohol and cigarette consumption in Russia during transition. Econ. Hum. Biol. 13, 128–143.

Holt, M.T., Goodwin, B.K., 1997. Generalized habit formation in an inverse almost ideal demand system: an application to meat expenditures in the US. Empirical Econ. 22 (2), 293–320.

Hood, H.B., Dorfman, J.H., 2015. Examining dynamically changing timber market linkages. Am. J. Agric. Econ. 97 (5), 1451–1463.

Kikulwe, E.M., Birol, E., Wesseler, J., Falck-Zepeda, J., 2011. A latent class approach to investigating demand for genetically modified banana in Uganda. Agric. Econ. 42 (5), 547–560.

Lin, B., Dong, D., Carlson, A., Rahkovsky, I., 2017. Potential dietary outcomes of changing relative prices of healthy and less healthy foods: the case of ready-to-eat breakfast cereals. Food Policy 68, 77–88.

Lin, B., Smith, T.A., Lee, J., Hall, K.D., 2011. Measuring weight outcomes for obesity intervention strategies: the case of a sugar-sweetened beverage tax. Econ. Hum. Biol. 9 (4), 329–341.

Lundbergh, S., Teräsvirta, T., Van Dijk, D., 2003. Time-varying smooth transition autoregressive models. J. Bus. Econ. Stat. 21 (1), 104–121.

New York State Department of Health, 2010. Summary of Tax on Sugar-Sweetened Beverages (Beverage Syrups and Soft Drinks). NYSDOH, Albany.

Omay, T., Van Eyden, R., Gupta, R., 2018. Inflation–growth nexus: evidence from a pooled CCE multiple-regime panel smooth transition model. Empirical Econ. 54 (3), 913–944.

Ortega, D.L., Wang, H.H., Wu, L., Olynk, N.J., 2011. Modeling heterogeneity in consumer preferences for select food safety attributes in china. Food Policy 36 (2), 318–324.

Pacula, R.L., 1998. Does increasing the beer tax reduce marijuana consumption? J. Health Econ. 17 (5), 557–585.

Pittman, G.F., 2005. Drivers of Demand, Interrelationships, and Nutritional Impacts within the Nonalcoholic Beverage Complex (PhD diss.). A&M University, Texas.

Rudd Center for Food Policy and Obesity, 2010. Revenue Calculator for Soft Drink Taxes. <http://www.yaleruddcenter.org/sodatax.aspx>.

Seleteng, M., Bittencourt, M., Van Eyden, R., 2013. Non-linearities in inflation–growth nexus in the SADC region: a panel smooth transition regression approach. Econ. Model. 30, 149–156.

Wang, E.Y., 2015. The impact of soda taxes on consumer welfare: implications of storability and taste heterogeneity. Rand J. Econ. 46 (2), 409–441.

Zhen, C., Finkelstein, E.A., Nonnemaker, J.M., Karns, S.A., Todd, J.E., 2014. Predicting the effects of sugar-sweetened beverage taxes on food and beverage demand in a large demand system. Am. J. Agric. Econ. 96 (1), 1–25.

Zhen, C., Wohlgenant, M.K., Karns, S., Kaufman, P., 2011. Habit formation and demand for sugar-sweetened beverages. Am. J. Agric. Econ. 93 (1), 175–193.

Zheng, Y., Zhen, C., Dench, D., Nonnemaker, J.M., 2017. US demand for tobacco products in a system framework. Health Econ. 26 (8), 1067–1086.